

Nuthan Reddy

Lead Data & Software Engineer | Data Governance | Distributed Systems

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SUMMARY

Lead Software Engineer with 10+ years building large-scale data platforms and governance systems across Azure, AWS, and GCP. Strong hands-on expertise in Python/Java, Apache Spark, and modern Data Lake architectures. Proven record owning end-to-end SDLC, driving design reviews, improving reliability with automated quality gates, and mentoring teams to deliver trusted, compliant data products.

WORK EXPERIENCE

Microsoft

Mar 2022 – Present

Senior Software Engineer

Hyderabad

- Led end-to-end SDLC for scalable data catalog for asset discovery, deletion and compliance tagging across 28B+ assets and 64 asset types in Microsoft.
- Designed the generic Facet Processing Framework for different Privacy Programs. Developed the Privacy Signals program for data deletion, export delivery, processing, and compliance reporting at enterprise scale.
- Architected Agent based Investigation and Remediation framework for incidents, reducing mean time to resolution by almost half and in some cases this entirely eliminated the need for manual intervention.
- Redesigned distributed scan bottlenecks to deliver 2-20x performance gains, improving data freshness from 8 days to 24 hours.
- Defined and enforced Data Contracts across DataMap, DataGrid, and Trust360, reducing live-site incidents by 50% and improving recovery time by 80%.

Skills: Data Governance, Agentic Systems, Azure Foundry, Python, Apache Spark, Databricks, Azure Data Factory, Synapse, C#, Data Lake, Delta, CI/CD

Walmart

Oct 2020 – Mar 2022

Software Engineer III

Bangalore

- Mentored junior engineers and improved team effectiveness through design discussions and distributed systems coaching.
- Contributed to a modular, rule-based, config-driven data platform for Enterprise Business Services teams.
- Optimized Spark data processing jobs, reducing execution time by ~30% and improving resource utilization.
- Built a Go-based Code Freeze app integrating Slack and GitHub to automate deployment governance workflows.

Skills: Apache Spark, Scala, GCP, BigQuery, Go, Slack, GitHub

Sigmoid

Jan 2019 – Oct 2020

Software Development Engineer I

Bangalore

- Sole developer of a planet-scale OpenX Prebid streaming pipeline supporting 100k req/sec and 1GB/sec throughput, presented at ApacheCon NA 2020.
- Improved reliability and reduced costs by implementing circuit-breaker patterns and optimizing resource usage.
- Led BigQuery-based NitroDB implementation, achieving ~30% cost reduction while maintaining performance.
- Built multi-tenant discovery/search capabilities with Elasticsearch for high-speed data exploration.
- Contributed critical bug fixes to the official spark-snowflake connector, improving data correctness.
- Implemented sieve-based query optimization features in NitroDB to improve query efficiency.

Skills: Kafka, AWS, Glue, EMR, Java, S3, Kubernetes, Docker, BigQuery, Elasticsearch

Capgemini

Jun 2015 – Jan 2019

Consultant P2

Bangalore

- Migrated business logic from Teradata to Hive for a leading energy client and optimized workloads using partitioning, file formats, and cost-based optimization.
- Built a real-time sensor analytics pipeline using Spark Streaming, NiFi, HBase and Power BI to predict oil well failures.
- Developed custom Java UDFs to optimize complex transformations and close platform feature gaps.

Skills: Hive, Apache Spark, NiFi, HBase, Java, Teradata

EDUCATION

Indian Institute of Technology (IIT) Kharagpur <i>Bachelor of Technology, Industrial & Systems Engineering</i>	2011 – 2015
Sri Chaitanya Jr College, Gudavalli <i>XII, Board of Intermediate Education</i>	2009 – 2011
Gowtham Concept School, Gudivada <i>X, Board of Secondary Education</i>	2008 – 2009

SKILLS

Core Technologies : Apache Spark, Distributed Systems, Data Modeling, RDBMS, Elasticsearch
Data Platforms : Data Lake, Lakehouse, Delta Lake, Parquet, BigQuery, Snowflake
Languages : Python, Java, Scala, SQL, Go, Shell Scripting
Engineering Excellence : Git, CI/CD, Automated Testing, Agile/Scrum, System Design, Code Reviews
Cloud & Infrastructure : Azure, AWS, GCP, Databricks, Docker, Kubernetes
Data Governance : Data Governance, Data Quality, GDPR Compliance

CERTIFICATIONS

Databricks Fundamentals	2026
Cloud Computing and Distributed Systems	2023
AWS Certified Solutions Architect – Associate	2022

COURSES

Agentic AI with Azure AI Foundry - <i>Microsoft Learn</i>	2025
Data Structures and Algorithms - <i>IIT Kharagpur</i>	2015
Soft Computing and Heuristic Search - <i>IIT Kharagpur</i>	2015
Java Memory Management - <i>LinkedIn Learning</i>	
Google Cloud Platform Big Data and Machine Learning Fundamentals - <i>Coursera</i>	
Apache Kafka Series - Kafka Cluster Setup & Administration - <i>Udemy Academy</i>	
Feature Selection for Machine Learning - <i>Udemy Academy</i>	
AWS Certified Developer - Associate - <i>Udemy</i>	2018
AWS Certified Solutions Architect - Associate - <i>Udemy</i>	2018
Python for Data Science and Machine Learning Bootcamp - <i>Udemy Academy</i>	
Cloud Computing and Distributed Systems - <i>IIT Patna</i>	